

might not solve the problem or achieve the desired effect. However, this was normally not an issue of lack of disclosure, but rather of claim construction. Whether the claims, description and figures provided the skilled person with sufficient information to carry out an invention was a purely technical question that was separate from that of what reasonably fell within the ambit of the claim wording. In the board's view, if the skilled person, upon consideration of the entire disclosure and possibly using common general knowledge, could **infer what would and what would not work**, a claimed invention was sufficiently disclosed, even if a broad construction might also encompass what did not work. In the case in hand, the skilled person would be able to directly recognise and exclude embodiments which obviously would not achieve the effect sought. In particular, the skilled person would disregard an upper part placed much lower than 30 metres above sea level. The board thus confirmed the opposition division's positive assessment of sufficiency (Art. 100(b) EPC).

In T 1583/17 the invention concerned the use of coated films. The board stated that it was important to note that there was a difference between whether or not a product met a **pre-set quality requirement** (here: the absence of glossy spots) and whether or not it met the requirements set out in a claim. That the quality of the coatings obtained did not satisfy the intended quality standard did not mean that those coatings did not satisfy the requirements of claim 1, which was the only relevant point when discussing sufficiency of disclosure. The alleged lack of enablement over the entire range had not been shown.

See also in this chapter II.C.7.1.2.

Objection of insufficient disclosure based on the subject-matter of a dependent claim

In T 2007/16 the opposition division had referred to what it called the "main rule" for Art. 83 EPC, according to which only one way to perform the invention needed to be clear to the skilled person. It had argued that a patent could not conflict with Art. 83 EPC if those skilled in the art could perform the subject-matter of an independent claim, irrespective of whether they could perform the subject-matter of a dependent claim. The board could not endorse this approach; the "main rule" asserted by the opposition division had no basis in the EPC or in the case law of the boards of appeal. A dependent claim directed at a particular embodiment could give rise to an objection under Art. 83 EPC if the skilled person did not know how to obtain this embodiment. Also, a dependent claim directed at subject-matter that the skilled person would not know how to obtain indicated that the invention defined in the corresponding independent claim was not sufficiently disclosed over the whole scope encompassed by the claim.

5.5. Parameters

If an essential feature of the invention is expressed by a parametric definition, the question is whether the parameter is so defined that the person of the art, on the basis of the disclosure of the patent as a whole and using his common general knowledge, could identify, without undue burden, the technical measures leading to the claimed subject-matter (T 61/14).

This section looks at the case law on parameters, which often addresses them in the context of methods of measuring them. Parameters are also dealt with elsewhere in this chapter: II.C.6.6.4 "Forbidden area of the claims" and II.C.8.2 "Article 83 EPC and clarity of claims".

5.5.1 Ambiguous parameters

a) Essential parameters

In T.815/07 (absorbent article defined by reference to a parameter range to be measured by a specified method) the board pointed out (Catchword) that the purpose of a parameter contained in a claim is to define an essential technical feature of the invention. Its significance is that the presence of this technical feature contributes to the solution of the technical problem underlying the invention. The method specified for determining the parameter should therefore be such as to produce **consistent values**, so that the skilled person will know when he carries out the invention whether what he produces will solve the problem or not. This decision was cited (and endorsed) in T.120/08, T.593/09 and T.1305/15. According to T.593/09 ("ill-defined ("unclear", "ambiguous") parameter"), the assertion that the skilled person is unable to determine whether he is working within the claimed scope as such is not a valid basis for denying sufficiency of disclosure (see point 4.1.4 of the Reasons). What is decisive is whether the parameter is so ill-defined that the skilled person is not able, on the basis of the disclosure as a whole and using his common general knowledge, to identify (without undue burden) the technical measures (e.g. selection of suitable compounds) necessary to solve the problem underlying the patent at issue. The board in T.1845/14 (see below) disagreed with some of the findings made in these decisions. Neither T.256/87 (point 17 of the Reasons) nor T.815/07 (point 6 of the Reasons) was followed by T.1960/14 in relation to the issue whether the skilled person would know whether he was working within or outside the scope of the claim.

In T.59/18 the relaxation ratio was an essential feature affecting the final properties of the obtained film and, thus, was relevant for solving the technical problem addressed in the patent. Therefore in accordance with the decisions T.593/09 and T.2403/11, the lack of clarity of the term "relaxation ratio" necessarily affected the sufficiency of disclosure since it was not possible to realize the process for preparing the claimed multilayer film without knowing how to achieve, to measure and to control this ratio.

In T.147/12 the board held that the conclusion reached in T.815/07 could not be applied to the present case. In T.815/07, claim 1 had pertained to an absorbent article comprising an absorbent core and a crotch region defined by its absorbent capacity relative to the absorbent core's total absorbent capacity, the absorbent capacity being determined by test method A. The board had found that the structure of the article claimed would have an effect on the actual amount of liquid absorbed. Because the patent did not provide adequate information about the tested articles as regards their structure, materials or regions, the test method defined in claim 1 resulted in totally arbitrary values for the crotch region's absorbent capacity. The claimed subject matter was therefore not sufficiently disclosed. In T.147/12, however, the alkali metal content only depended on its method of

determination; it did not depend on any variable or feature characterising the process claimed.

In T 808/09 (cartridge for use in a beverage preparation machine / liquid chocolate ingredient), the board considered that the patent failed to enable the measurement of the essential parameter of the invention, i.e. the viscosity of the liquid chocolate ingredient. Even if this viscosity feature had been shifted into a preamble of method claim 1 and supposing this preamble had been directed to the prior art, this shifting of the feature would still not have solved the problem of insufficiency. For a reference to prior art in the preamble to adequately serve this purpose the patent in suit still needs to contain all the necessary information in sufficient detail in order to enable the person skilled in the art to perform the invention. In the case before the board this would require information as to the viscosity measurement device to be used and the parameters to be observed. All this information was missing.

T 1845/14 decided that in case of an unclear parameter defined in a claim whose values required in the claim are indicated in the specification to be essential to solving the problem underlying the patent at issue, the ability of the skilled person to solve that problem by reproducing what is claimed is **not a suitable criterion** for assessing sufficiency of disclosure when the problem or an effect derivable from it are not explicitly or implicitly part of the definition of the claimed subject-matter (point 9.8 of the Reasons). T 1845/14 addressed the rationale of T 593/09, T 815/07, T 172/99 and T 608/07 ("promise of the invention" is rather an issue of inventive step) and stated that all the decisions cited above relating to a parameter mentioned in a claim whose definition was ambiguous are based on a definition of the term "invention" which does not refer to the combination of features defined by the terms of the claim under consideration but rather to the inventive idea or concept the inventor had in mind that led the patentee to seek protection for what is claimed. The board in T 1845/14, making detailed reference in its reasons to landmark decisions such as T 435/91 (OJ 1995, 188) and T 939/92 (OJ 1996, 309), was convinced that, provided that the problem indicated in the patent in suit was not implied by the wording of the claims in terms of, for example, an effect or a functional definition, there was no apparent justification to deny a contribution to the art justifying a patent monopoly merely because the claimed subject-matter could not be considered to solve a problem initially indicated in the description. The term "invention" corresponds, in accordance with R. 43(1) EPC, to the specific combination of features in the claim, as was reminded in G 2/98 (OJ 2001, 413), whose definition is used when issues of priority, novelty and inventive step are considered. The board in T 1845/14 did not see any reason to consider a different meaning of the term "invention" in relation to the issue of sufficiency of disclosure. There is no reason to define the invention on the basis of an effect alleged in the patent to be achieved by the claimed subject-matter or on the basis of specific conditions for measuring a parameter when the wording of the claim does not require this. The invention is defined by the terms of the claims, which should be given their broadest technical sensible meaning in the context they appear. The board said that its position was also supported by G 1/03 (OJ 2004, 413 – concerning non-working embodiments).

The decision in T 1845/14 was endorsed on this point by the subsequent case law in T 1900/17, T 409/17, T 1260/16, T 189/16, T 1744/14. No decision taking issue with

T.1845/14 was found. In T.786/15 the board decided, in view of the possibility of calibrating the method of measuring the T_g parameter by reverse engineering (parameter not ambiguous), that the case law on ambiguous parameters (in particular T.1845/14) did not apply.

Other decisions concerning the measurement of essential parameters: T.83/01 (specific mean diameter), T.1250/01 (Sears number), T.808/09 (viscosity).

b) Several known methods of determining a parameter

In both T.485/00 and T.225/93, three methods were known in the art for the determination of the specific surface area of a CaCO₃ particle. In neither case did the description or common general knowledge indicate a preference for one of them. In T.485/00, the board held that reproducing an example and measuring the surface area of the resulting product by two or three well-known methods did not represent an undue burden for the skilled person. In T.225/93, however, the board found that, as there were three different measuring methods which did not always lead to the same result, this amounted to an **undue burden**.

T.473/15 observed that the conclusions of T.225/93 could not be applied since in that case the principles underlying the possible methods of measurement were completely different from each other (permeability, photometry and adsorption), contrary to the case at hand, and in that case the parties agreed that the different methods yielded different results. On the contrary, in case T.473/15 the appellants (opponents) did not provide evidence for the presence of major deviations between the different definitions for fiber diameter.

In T.147/12, the objection of the appellant (opponent) was not that no method of determination of the alkali metal content in polyethers existed. Rather, the gist of its argumentation was that D7 (scientific publication), D8 (a study) and D11 (experimental report provided by the opponent) showed that the value obtained for the alkali metal content in the polyether depended on the analytical method used for its determination. The board stated that even if the conditions of measurement lead to **variations** of the value of the alkali metal content as argued by the appellant, this **alone did not** constitute a lack of sufficiency of disclosure regarding the claimed subject-matter as a whole since it had not been shown that the uncertainty concerning the alkali metal content affected the claimed process to such an extent that the skilled person wishing to perform the process would face an undue burden. The appellant had shown that the uncertainty concerning the method of determination of the alkali metal content meant that the skilled person could not ascertain whether the value he would obtain was within or outside the claimed range. It was, however, not shown that as a result of that uncertainty, the skilled person would fundamentally be prevented from obtaining a polyether according to claim 1. The board stated that T.83/01 (skilled person not in a position to perform any measurement of the claimed parameter) and T.815/07 (test method defined in claim 1 resulting in totally arbitrary values) were not applicable to the present case.

In T.345/16 the board considered that measuring a "particle size", a conventional parameter in the technical field concerned, would merely require selecting and applying

any of the well-known methods used in the field, for which no undue burden or inventive skill would be required. The appellant (opponent) argued that the patent in suit did not include any information as to how the "particle size" should be measured, and that this would lead to significant inconsistencies in the results. The board was not convinced by this reasoning, since any possible inconsistency resulting from the choice of a particular measuring method (among a number of well-known available alternatives) would only affect the demarcation of the scope of protection and not the ability to carry out the invention. In this respect the board noted that, in line with a number of decisions in the case law (see e.g. T 378/11, dealing precisely with the parameter "particle size"), problems of demarcation had to be dealt with under Art. 84 EPC and not under Art. 83 EPC. Since the feature "particle size" was part of the claims as granted, the board was not competent to assess compliance with Art. 84 EPC (G 3/14). The board also noted that there was no requirement in claim 1 to measure the primary particle size and the agglomerate particle size simultaneously rather than sequentially. Furthermore, a number of well-known optical methods were available which would allow both a simultaneous (e.g. visual analysis of the agglomerated particles to estimate the size of the agglomerates and the primary particles identifiable in these agglomerates) and a sequential measurement to be carried out. In any case, the appellant had not provided any evidence in support of its allegations.

In T 1154/12 the respondent (opponent) contended that the patent lacked sufficient disclosure because it did not define the method which should be used for measuring and calculating the "average particle size" called for in claim 1. The board was not persuaded by this argument. It was undisputed that several methods of measuring or calculating the average particle size were conceivable to the skilled person. Whether or not the use of several kinds of measuring or calculating the average particle size led to different results was, however, a matter of determining the boundaries of the independent claim, i.e. a matter of clarity rather than sufficiency of disclosure (see T 378/11).

In T 2666/17 confirmed and summarised the findings in T 815/17 (problem if the parameter is ill defined) and noted that surface tension was a well-known parameter which could be measured using known methods. That the measuring method would lead to different results would not prevent the skilled person from reproducing the invention but would simply raise doubts as to whether certain embodiments at the margins of the scope of the protection fall within the forbidden area or not (issue of Art. 84 EPC).

In T 1960/14 (parameter – melting point of a palm oil fraction), the board stated that the skilled person (common general knowledge) was aware of three suitable standard methods. The respondent (opponent) asserted that these methods provided significantly different results but did not submit any experimental evidence. In view of tests disclosed in D22, the board concluded in agreement with the patent proprietor that all three methods provided very similar results with uncertainty only at the edges of the claimed ranges. T 1960/14 endorsed the finding in T 608/07 that for an insufficiency arising out of ambiguity it is not enough to show that an ambiguity exists, in this case at the margins of the claimed melting point range due to the lack of any indication of the measuring method. It will normally be necessary to show that this ambiguity deprived the skilled person of the promise of the invention. In the case at issue, the respondent (opponent) had not submitted any technical evidence to that effect. The respondent in T 1960/14 also referred